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/Users/mehakagrawal/PycharmProjects/Final_Dissertation/.venv/bin/python /Users/mehakagrawal/Desktop/Final_Dissertation/Codes/4.tert_3month_de_analysis/tert_3month_de_analysis.py
STEP 1: Building combined count matrix from 13 featureCounts files
Found 13 featureCounts files
  M3_WT_234: 36.1M reads mapped
  M3_Het_235: 14.9M reads mapped
  M3_Het_251: 36.4M reads mapped
  M3_WT_253: 37.0M reads mapped
  M3_MUT_254: 29.1M reads mapped
  M3_WT_255: 34.9M reads mapped
  M3_MUT_256: 43.7M reads mapped
  M3_WT_257: 41.1M reads mapped
  M3_Het_258: 27.7M reads mapped
  M3_Het_264: 47.5M reads mapped
  M3_WT_270: 10.6M reads mapped
  M3_Het_271: 36.7M reads mapped
  M3_MUT_273: 44.1M reads mapped

Combined count matrix: 32105 genes x 13 samples
Saved: tert3m_raw_counts.csv


STEP 2: Metadata
           animal_id genotype  age
M3_WT_234        234       WT  3mo
M3_Het_235       235      Het  3mo
M3_Het_251       251      Het  3mo
M3_WT_253        253       WT  3mo
M3_MUT_254       254      MUT  3mo
M3_WT_255        255       WT  3mo
M3_MUT_256       256      MUT  3mo
M3_WT_257        257       WT  3mo
M3_Het_258       258      Het  3mo
M3_Het_264       264      Het  3mo
M3_WT_270        270       WT  3mo
M3_Het_271       271      Het  3mo
M3_MUT_273       273      MUT  3mo

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Groups: {'WT': 5, 'Het': 5, 'MUT': 3}


STEP 3: Pre-filtering
Before: 32105 genes
After (>= 10 reads in >= 3 samples): 18522 genes
Removed: 13583


STEP 4: Normalisation + PCA
PCA variance: PC1=53.2%, PC2=8.5%, PC3=7.8%


DE Analysis: A: MUT vs WT (3mo)
Using None as control genes, passed at DeseqDataSet initialization
Fitting size factors...
... done in 0.00 seconds.

Fitting dispersions...
... done in 1.77 seconds.

Fitting dispersion trend curve...
... done in 0.20 seconds.

Fitting MAP dispersions...
... done in 2.27 seconds.

Fitting LFCs...
... done in 1.44 seconds.

Calculating cook's distance...
... done in 0.01 seconds.

Replacing 0 outlier genes.

Running Wald tests...
... done in 0.44 seconds.

Log2 fold change & Wald test p-value: genotype MUT vs WT
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                         baseMean  log2FoldChange  ...    pvalue      padj
Geneid                                                ...

ENSDARG00000104632   27.385337       -0.397447  ...  0.445041  0.999855
ENSDARG00000100660  280.199307        0.219747  ...  0.599506  0.999855
ENSDARG00000100422   45.063546        1.155630  ...       NaN       NaN
ENSDARG00000102128  128.521303        0.085558  ...  0.868399  0.999855
ENSDARG00000103095  274.560056        0.319052  ...  0.251135  0.999855
...                        ...             ...  ...       ...       ...
ENSDARG00000101185   50.864420        1.275981  ...  0.000045  0.110347
ENSDARG00000102248    8.897637       -0.191274  ...  0.784828  0.999855
ENSDARG00000102790   77.325523       -0.340791  ...  0.507792  0.999855
ENSDARG00000100589   20.453289        0.467207  ...  0.485225  0.999855
ENSDARG00000099551    5.872410       -0.536178  ...  0.501458  0.999855

[18522 rows x 6 columns]

Significant DEGs (q<0.05, |log2FC|>1.0): 0   [pi0=1.000]
  Up: 0
  Down: 0

Top 15 by q-value:


DE Analysis: B: Het vs WT (3mo)
Using None as control genes, passed at DeseqDataSet initialization
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```
Fitting size factors...
... done in 0.01 seconds.

Fitting dispersions...
... done in 1.68 seconds.

Fitting dispersion trend curve...
... done in 0.21 seconds.

Fitting MAP dispersions...
... done in 1.87 seconds.

Fitting LFCs...
... done in 1.18 seconds.

Calculating cook's distance...
... done in 0.01 seconds.

Replacing 0 outlier genes.

Running Wald tests...
... done in 0.44 seconds.

Log2 fold change & Wald test p-value: genotype Het vs WT
                      baseMean  log2FoldChange  ...    pvalue      padj
Geneid                                          ...
ENSDARG00000104632   28.352465        0.108583  ...  0.817364  0.999554
ENSDARG00000100660  234.142437       -0.059943  ...  0.866556  0.999554
ENSDARG00000100422   33.562607        0.517677  ...  0.194404  0.999554
ENSDARG00000102128  111.447328       -0.059558  ...  0.897893  0.999554
ENSDARG00000103095  241.631568        0.165133  ...  0.467310  0.999554
...                        ...             ...  ..
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.        ...        ...
ENSDARG00000101185   50.605337        1.252285   ...  0.010713  0.838098
ENSDARG00000102248    7.406850       -0.415049   ...  0.466543  0.999554
ENSDARG00000102790   85.138896        0.307937   ...  0.501776  0.999554
ENSDARG00000100589   20.005788        0.546829   ...  0.345844  0.999554
ENSDARG00000099551    7.372319        0.496200   ...  0.379861  0.999554

[18522 rows x 6 columns]

Significant DEGs (q<0.05, |log2FC|>1.0): 1   [pi0=1.000]
  Up: 1
  Down: 0

Top 15 by q-value:
  ENSDARG00000100265         log2FC=+4.43  qvalue=1.01e-02


DE Analysis: C: MUT vs Het (3mo)
Using None as control genes, passed at DeseqDataSet initialization
Fitting size factors...
... done in 0.00 seconds.

Fitting dispersions...
... done in 1.49 seconds.

Fitting dispersion trend curve...
... done in 0.19 seconds.

Fitting MAP dispersions...
... done in 2.10 seconds.

Fitting LFCs...
... done in 1.25 seconds.
```

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Calculating cook's distance...
... done in 0.01 seconds.

Replacing 0 outlier genes.

Running Wald tests...
... done in 0.52 seconds.

Log2 fold change & Wald test p-value: genotype MUT vs Het
                      baseMean  log2FoldChange  ...    pvalue      padj
Geneid                                                ...
ENSDARG00000104632   32.792484       -0.519580  ...  0.324605  0.999873
ENSDARG00000100660  306.605075        0.260979  ...  0.608838  0.999873
ENSDARG00000100422   58.589045        0.650598  ...       NaN       NaN
ENSDARG00000102128  140.807515        0.131192  ...  0.784478  0.999873
ENSDARG00000103095  330.862168        0.135403  ...  0.571798  0.999873
...                        ...             ...  ...       ...       ...
ENSDARG00000101185   88.958632        0.007795  ...  0.989797  0.999873
ENSDARG00000102248    8.188144        0.218954  ...  0.713514  0.999873
ENSDARG00000102790  101.616022       -0.668832  ...  0.150353  0.999873
ENSDARG00000100589   29.298148       -0.103236  ...  0.849233  0.999873
ENSDARG00000099551    8.826369       -1.070213  ...  0.090926  0.999873

[18522 rows x 6 columns]

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```
Significant DEGs (q<0.05, |log2FC|>1.0): 1   [pi0=1.000]
  Up: 0
  Down: 1

Top 15 by q-value:
  ENSDARG00000099419          log2FC=-5.46  qvalue=4.53e-04


STEP 6: Volcano plots
  Saved: volcano_MUT_vs_WT.png
  Saved: volcano_Het_vs_WT.png
  Saved: volcano_MUT_vs_Het.png


STEP 7: Cross-dataset overlap (3mo vs 18mo genotype effect)
Genes in 3mo dataset: 18522
Genes in 18mo dataset: 22253
Shared gene IDs: 16873
3mo-only genes: 1649
18mo-only genes: 5380

3mo MUT vs WT DEGs: 0
18mo MUT vs WT DEGs (by qvalue): 5541
DEG overlap: 0


SUMMARY

Dataset: 13 muscle samples, 3 months old zebrafish
  WT (tert +/+): n=5
  Het (tert +/-): n=5
  MUT (tert -/-): n=3

Genes analysed: 18522

Comparison A - MUT vs WT:
  DEGs: 0 (up:0 down:0)

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Comparison B - Het vs WT:
  DEGs: 1 (up:1 down:0)

Comparison C - MUT vs Het:
  DEGs: 1 (up:0 down:1)

PCA: PC1=53.2%, PC2=8.5%

All outputs saved to: /Users/mehakagrawal/Desktop/Final_Dissertation/Outputs/4. tert_3month_de_analysis

Process finished successfully!

Process finished with exit code 0

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